

MENSURATION-2D

1. The perimeters of three equilateral triangles are in the ratio of 6:4:1 respectively. What is the ratio of their respective areas?

A. 36 : 16 : 1

B. 6 : 4 : 1

C. 12 : 8 : 1

D. 3 : 2 : 1

2. If AO and BO are angle bisector of angle A and angle B respectively of a square ABCD with side 20 cm, find the area of the triangle AOB.

- a) 100 sq cm b) 150 sq cm c) 50 sq cm d) 200 sq cm**

3. The area enclosed between two concentric circles is 770 sq cm. The radius of the inner circle is 14 cm. What is the circumference (in cm) of the outer circle?

a) 132 b) 143 c) 110 d) 88

4. Area of a rhombus is 162 sq cm and length of one of its diagonal is equal to the side of equilateral triangle of area $36\sqrt{3}$ sq cm. What is the difference between the diagonals of the rhombus?

a) 15 cm

b) 9 cm

c) 27 cm

d) 12 cm

5. The perimeter of a rectangular field is 178m and the length of its diagonal is 65m. The cost of leveling the field at Rs. 8.50 per sq m is:

- a) Rs. 15708
- b) Rs. 13874
- c) Rs. 14784
- d) Rs. 15810

6. Find the circumference of circle, whose area is 24.64sq cm

a) 14.64 m b) 16.36 m c) 17.60 m d) 18.40 m

7. The base of a parallelogram is 27 units and the area of the isosceles triangle drawn on the same base between the parallel lines of the parallelogram is 648 square units. Find the vertical distance between the parallel lines.

- a) 48**
- b) 15**
- c) 19**
- d) 25**

8. The length of a wire is 42 cm. From this wire, a rectangle of length 12 cm is made. Find the ratio of the area of this rectangle to that of a circle of radius 7 cm.

- a) 54 : 77
- b) 11 : 23
- c) 23 : 33
- d) 27 : 38

9. A large rectangular park 500 m long and 300 m broad has two 2 m broad walkways — one parallel and the other perpendicular to the length. If this walkway is to be paved by tiles $\frac{1}{2}$ m $\times$ $\frac{1}{2}$ m, how many tiles would be required?

- a) 6384**
- b) 6400**
- c) 1596**
- d) 3192**

THE END